Summary: GATE AE 2026 Paper

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1 PNC- Number of 3 digit numbers

Q.4	How many 3-digit numbers can be formed using three distinct single digit prime numbers?
(A)	64
(B)	24
(C)	12
(D)	4

Problem Statement:
How many 3-digit numbers can be formed using three distinct single digit prime numbers?

(A) 64

(B) 24

(C) 12

(D) 4

Step-by-Step Solution

Step 1: There are 4 single digit prime numbers: 2,3,5 and 7. To make a three digit number, we choose 3 out of these and arrange them.

Step 2: Number of ways to choose three numbers = ${}^4C_3$ and number of way to arrange them = 3!

Step 3: By multiplication principle, total number of ways to achieve our requirement = ${}^4C_3 \times 3! = 4 \times 6 = 24$

2 Problems on Probability

2.1 Number of students in a class

Q.5	In a group of students, 10 students like Mathematics, 12 students like English, 4 students like both Mathematics and English, and 6 students like neither Mathematics nor English. The number of students in the group is _____
(A)	18
(B)	20
(C)	24
(D)	32

Problem statement:
In a group of students, 10 students like Mathematics, 12 students like English, 4 students like both Mathematics and English, and 6 students like neither Mathematics nor English. The number of students in the group is ______.

(A) 18

(B) 20

(C) 24

(D) 32

Step-By-Step Solution (Probability-Based Method):

Step 1: Let N be the total number of students in the group.
If one student is chosen uniformly at random from the group, define:

- M: The event that the student likes Mathematics.
- E: The event that the student likes English.

Step 2: The respective probabilities are expressed as:

$$P(M) = \frac{10}{N}$$

$$P(E) = \frac{12}{N}$$

$$P(ME) = \frac{4}{N} \quad \text{(student likes both)}$$

$$P((M \cup E)') = \frac{6}{N} \quad \text{(student likes neither)}$$

Step 3: Since the total probability of the sample space is 1:

$$P(M \cup E) + P((M \cup E)') = 1$$

Step 4: Applying the Addition Theorem of Probability:

$$P(M) + P(E) - P(ME) + P((M \cup E)') = 1$$

Step 5: Substitute the probabilities in terms of N:

$$\frac{10}{N} + \frac{12}{N} - \frac{4}{N} + \frac{6}{N} = 1$$

$$\frac{10 + 12 - 4 + 6}{N} = 1$$

$$\frac{24}{N} = 1 \implies N = 24$$

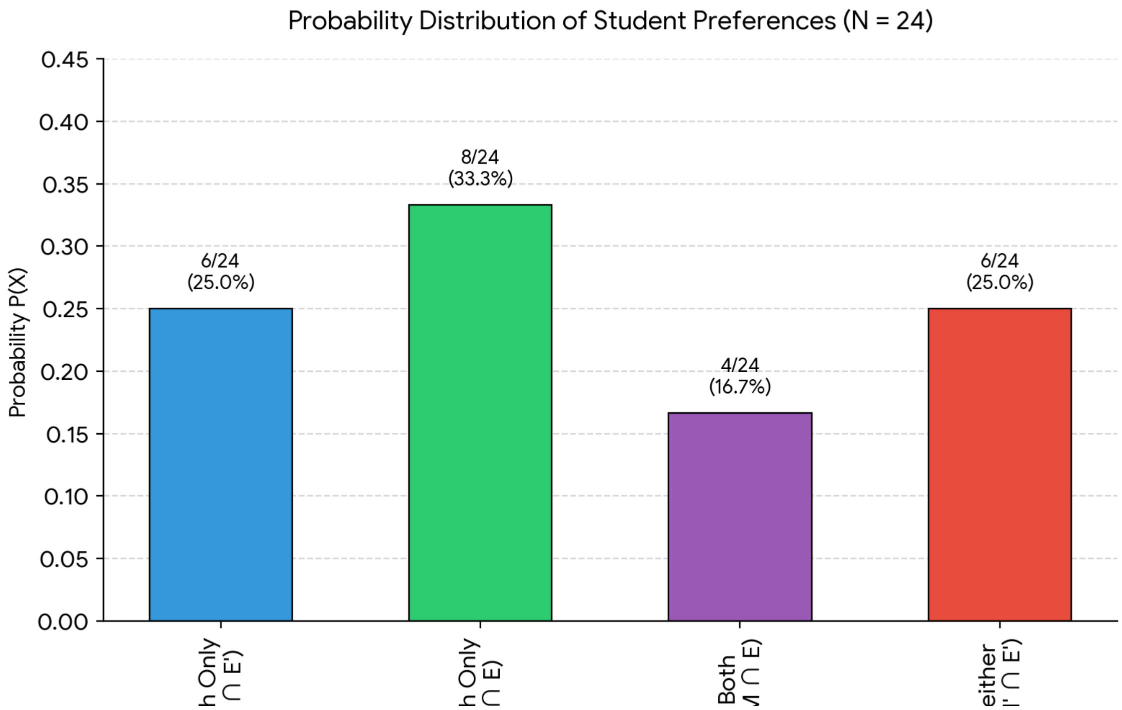
Correct Answer: (C) 24

Python Codes

Python Implementation

```python
import matplotlib.pyplot as plt
import pandas as pd


num_math = 10
num_english = 12
num_both = 4
num_neither = 6

count_math_only = num_math - num_both  # 6
count_english_only = num_english - num_both  # 8
count_both = num_both  # 4
count_neither = num_neither  # 6

total_students = (
    count_math_only + count_english_only + count_both + count_neither
)

joint_prob_matrix = pd.DataFrame(
    [
        [count_both / total_students, count_math_only / total_students],
```

```python
        [count_english_only / total_students, count_neither / total_students],
    ],
    index=["Likes English (E)", "Dislikes English (E')"],
    columns=["Likes Math (M)", "Dislikes Math (M')"],
)
joint_prob_matrix["Total P(E)"] = joint_prob_matrix.sum(axis=1)
joint_prob_matrix.loc["Total P(M)"] = joint_prob_matrix.sum(axis=0)

print("=" * 60)
print(f"TOTAL STUDENTS (N) = {total_students}")
print("=" * 60)
print("\n--- 2x2 Joint Probability Distribution Matrix ---\n")
print(joint_prob_matrix.round(4))

categories = [
    "Math Only (M and E')",
    "English Only (M' and E)",
    "Both Math and English (M and E)",
    "Neither (M' and E')",
]
counts = [count_math_only, count_english_only, count_both, count_neither]
probabilities = [c / total_students for c in counts]

dist_df = pd.DataFrame(
    {
        "Event Category": categories,
        "Student Count": counts,
        "Probability P(X)": probabilities,
        "Fraction": [
            f"{c}/{total_students}" for c in counts
        ],  # Exact fractions
    }
)
print("\n\n--- Discrete Probability Distribution Table ---\n")
print(dist_df.to_string(index=False))

plt.figure(figsize=(9, 5))
bars = plt.bar(
    categories, probabilities, color=["#3498db", "#2ecc71", "#9b59b6", "#e74c3c"]
)

plt.title(
    "Probability Distribution of Student Preferences (N = 24)",
    fontsize=14,
    pad=15,
)
plt.ylabel("Probability P(X)", fontsize=12)
plt.ylim(0, 0.45)
plt.grid(axis="y", linestyle="--", alpha=0.7)

for bar, count in zip(bars, counts):
    yval = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width() / 2.0,
        yval + 0.01,
        f"{count}/{total_students}\n({yval:.1%})",
        ha="center",
        va="bottom",
        fontsize=10,
        fontweight="bold",
    )
plt.tight_layout()
plt.show()
```



The probability distribution

2.2 Problem on manipulating probabilities

Problem statement:
Consider two events A and B in a sample space S. We know the following data:

$$P(A), \quad P(B), \quad P(AB), \quad P((A \cup B)')$$

Find the value of $P(AB')$ using the following formulae:

- $A + A' = 1, \quad AA' = 0$
- $P(1) = 1, \quad P(0) = 0 \implies P(A + A') = 1, \quad P(AA') = 0$
- $P(A + A') = P(A) + P(A')$

Step-By-Step Solution:

Step 1: Express event A using the Identity $B + B' = 1$
Using the given identity that for any event B, $B + B' = 1$:

$$A = A \cdot 1 = A(B + B')$$

Applying the distributive law:

$$A = AB + AB'$$

Step 2: Verify that AB and AB' are Mutually Exclusive
To apply the additive property of probability, we check the intersection of AB and AB':

$$(AB)(AB') = A \cdot A \cdot B \cdot B' = A \cdot (BB')$$

Since $BB' = 0$:

$$(AB)(AB') = A \cdot 0 = 0$$

Thus, the events AB and AB' are mutually exclusive.

Step 3: Apply the Probability Addition Rule
Since $A = AB + AB'$ and $(AB)(AB') = 0$, we apply $P(X + Y) = P(X) + P(Y)$:

$$P(A) = P(AB + AB') = P(AB) + P(AB')$$

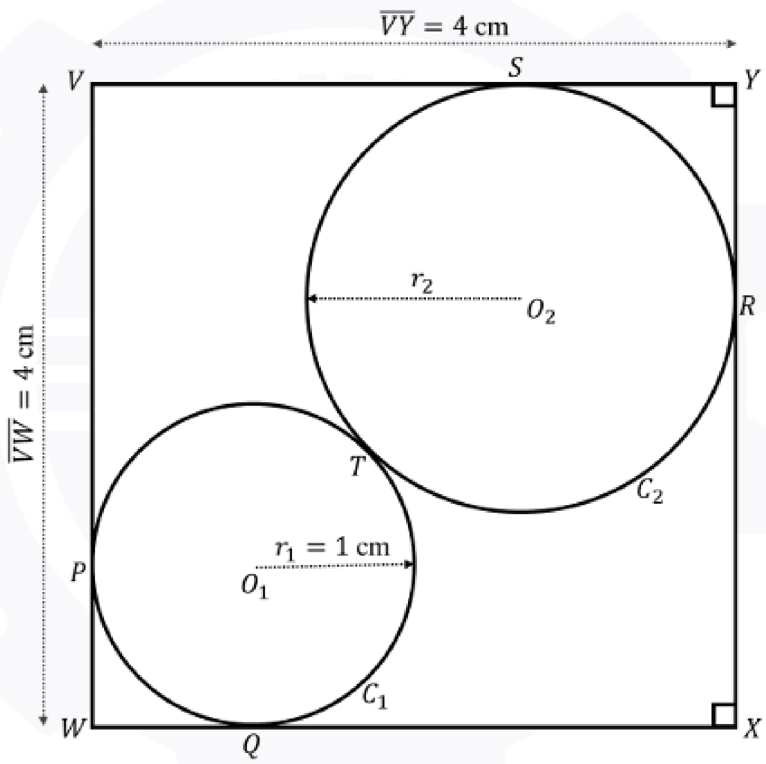
Step 4: Solve for $P(AB')$
Rearranging the equation from Step 3 to isolate $P(AB')$:

$$P(AB') = P(A) - P(AB)$$

Final Answer:

$$\mathbf{P(AB') = P(A) - P(AB)}$$

3 Touching square and circles

Q.10	As shown in the figure, circle C_1 with center O_1 and radius r_1 touches the square $VWXY$ at points P and Q while circle C_2 with center O_2 and radius r_2 touches the square $VWXY$ at points R and S. The two circles touch each other at T. Given $r_1 = 1$ cm and $\overline{VY} = \overline{VW} = 4$ cm, $r_2 =$ ______ cm. 
(A)	$4 - 3\sqrt{2}$
(B)	$1 + 2\sqrt{2}$
(C)	$7 - 4\sqrt{2}$
(D)	$5 + 3\sqrt{2}$

Problem Statement:
Given a square of side s and two circles, onw with known radius r_1 and another with unkown radius r_2, such that both of them touch each other externally and also the square, find the value of r_2. Also describe how the figure in the problem can be drawn.

Step-by-Step Solution:

Step 1: Let the side of the square be s, radius of the first circle be r_1 and that of the unknown circle be r_2.

Step 2: Clearly from figure, $\overline{WO_1}$ + $\overline{O_1T}$ + $\overline{TO_2}$ +$\overline{O_2Y}$ = $\sqrt{2}s$.

Step 3: We know that $\overline{WO_1}$ = $\sqrt{2}r_1$, $\overline{O_1T}$ = r_1, $\overline{TO_2}$ = r_2 and $\overline{O_2W}$ = $\sqrt{2}r_2$.

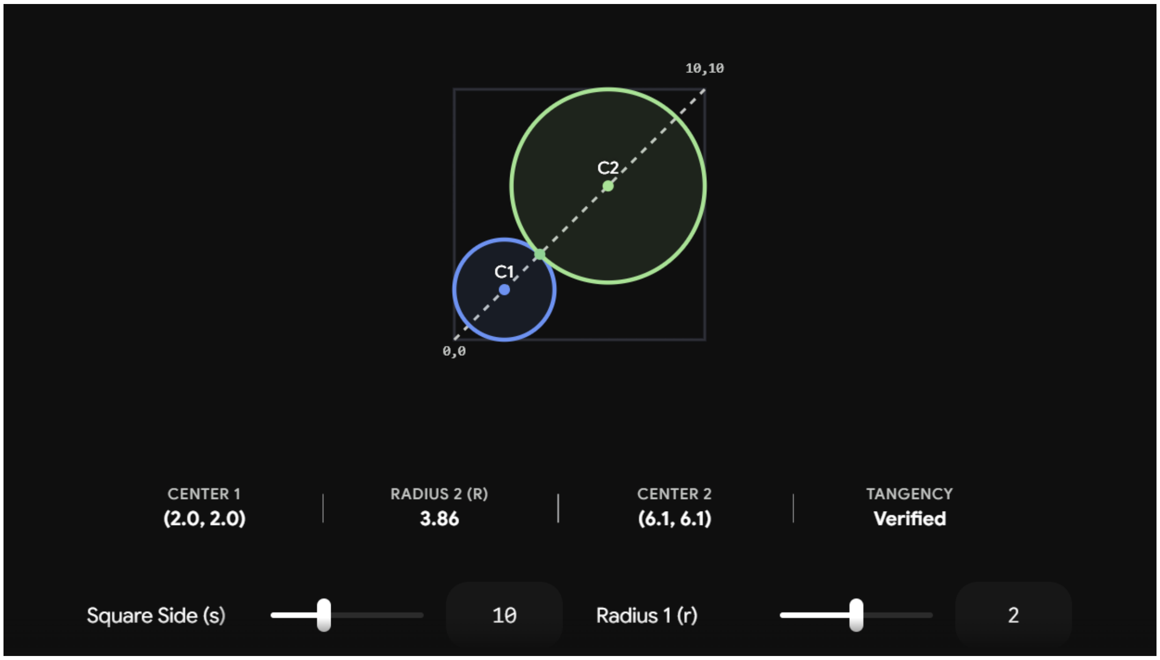
Step 4: Substituting this in our result from point 2, we get $r_2 = \frac{\sqrt{2}s-(\sqrt{2}+1)r_1}{\sqrt{2}+1}$.

Step 5: Now to draw the figure, we assign coordinates to vertices of the square(say (0,0), (s,0), (s,s) and (s,0), the centre of the first circle (r_1,r_1) and the centre of the third circle($s - r_2$,$s - r_2$) to get the following plot.

Python Codes

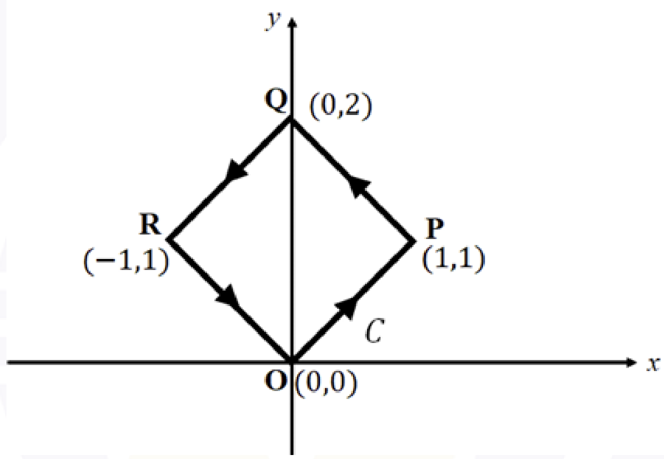
Python Implementation

```
import matplotlib.pyplot as plt
import matplotlib.patches as patches

# Define variables (ensure r < s)
s = 10.0  # Side length of the square
r = 3.0   # Radius of the circle

fig, ax = plt.subplots(figsize=(6, 6))

# 1. Draw Square: Bottom-left at (0,0) with width=s and height=s
# Vertices are (0,0), (0,s), (s,0), and (s,s)
square = patches.Rectangle((0, 0), s, s, fill=False, edgecolor='blue', linewidth=2, label=f'Square (s={s})')
ax.add_patch(square)

# 2. Draw Circle: Center at (r, r) with radius r
circle = patches.Circle((r, r), r, fill=True, facecolor='lightblue', edgecolor='red', alpha=0.6, label=f'Circle (r={r})')
ax.add_patch(circle)

# Mark key points
ax.plot([0, 0, s, s], [0, s, 0, s], 'bo', label='Vertices')  # Square corners
ax.plot(r, r, 'ro', label=f'Center ({r},{r})')               # Circle center

# Format canvas display
margin = s * 0.15
ax.set_xlim(-margin, s + margin)
ax.set_ylim(-margin, s + margin)
ax.set_aspect('equal')  # Keeps aspect ratio 1:1 so circle and square don't distort
plt.grid(True, linestyle='--', alpha=0.5)
plt.title(f"Square [0, {s}] with Circle Centered at ({r}, {r})")
plt.legend(loc='upper right')

plt.show()
```



The figure plotted

4 Problem on line integrals

Q.11	Consider the contour C shown in the figure below. For the vector $\vec{F} = (x + 2y)\hat{e}_x + (2x + 4y)\hat{e}_y$, the integral $\oint_C \vec{F} \cdot d\vec{l} =$_________. Here $d\vec{l}$ represents an infinitesimal length along the contour C. 
(A)	0
(B)	2
(C)	4
(D)	6

We can break the closed path C into its four straight line paths: $O \to P \to Q \to R \to O$.

The general line integral expression to evaluate along any path is:

$$I = \int (x + 2y)\, dx + (2x + 4y)\, dy$$

1. Path $O(0,0) \to P(1,1)$

- **Equation of line:** $y = x \implies dy = dx$
- **Limits:** x goes from 0 to 1

$$I_1 = \int_0^1 (x + 2x)\,dx + (2x + 4x)\,dx = \int_0^1 9x\,dx = \left[\frac{9x^2}{2}\right]_0^1 = \frac{9}{2}$$

2. Path $P(1,1) \to Q(0,2)$

- **Equation of line:** $y = 2 - x \implies dy = -dx$
- **Limits:** x goes from 1 to 0

$$I_2 = \int_1^0 (x + 2(2 - x))\,dx + (2x + 4(2 - x))(-dx)$$

$$I_2 = \int_1^0 [(4 - x) - (8 - 2x)]\,dx = \int_1^0 (x - 4)\,dx = \left[\frac{x^2}{2} - 4x\right]_1^0 = \frac{7}{2}$$

3. Path $Q(0,2) \to R(-1,1)$

- **Equation of line:** $y = x + 2 \implies dy = dx$
- **Limits:** x goes from 0 to -1

$$I_3 = \int_0^{-1} (x + 2(x + 2))\,dx + (2x + 4(x + 2))\,dx = \int_0^{-1} (9x + 12)\,dx$$

$$I_3 = \left[\frac{9x^2}{2} + 12x\right]_0^{-1} = \left(\frac{9}{2} - 12\right) - 0 = -\frac{15}{2}$$

4. Path $R(-1,1) \to O(0,0)$

- **Equation of line:** $y = -x \implies dy = -dx$
- **Limits:** x goes from -1 to 0

$$I_4 = \int_{-1}^0 (x + 2(-x))\,dx + (2x + 4(-x))(-dx) = \int_{-1}^0 x\,dx$$

$$I_4 = \left[\frac{x^2}{2}\right]_{-1}^0 = 0 - \frac{1}{2} = -\frac{1}{2}$$

Total Line Integral

Summing the integrals across all four line segments:

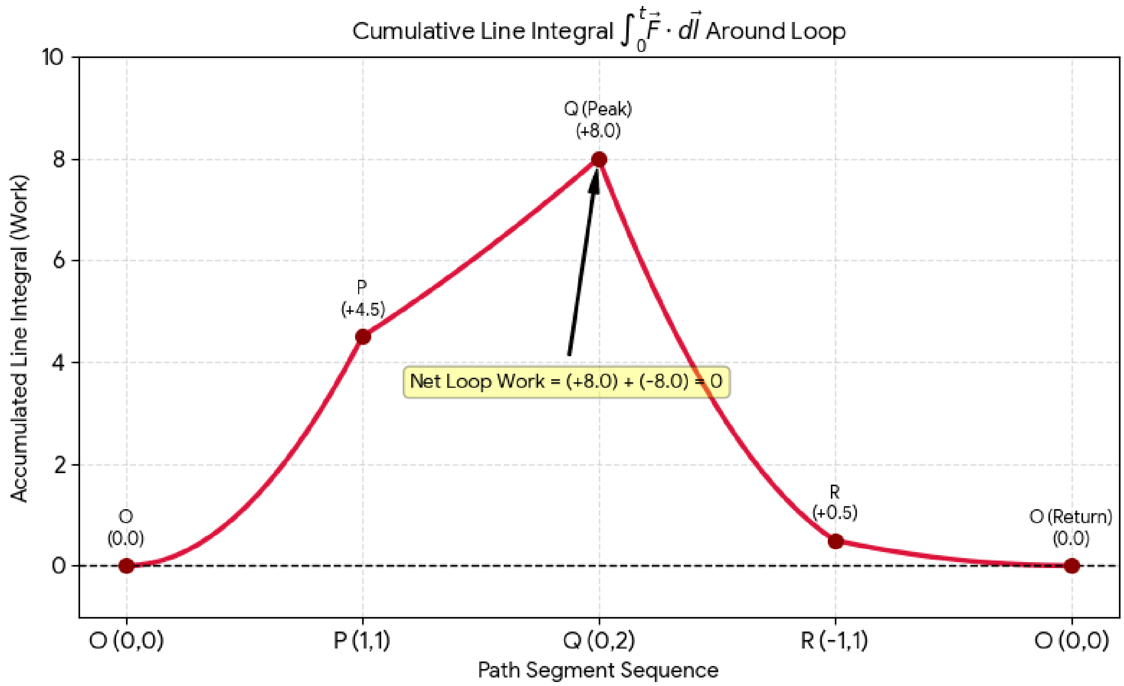
$$I = I_1 + I_2 + I_3 + I_4$$

$$I = \frac{9}{2} + \frac{7}{2} - \frac{15}{2} - \frac{1}{2} = \frac{16 - 16}{2} = 0$$

Python Codes

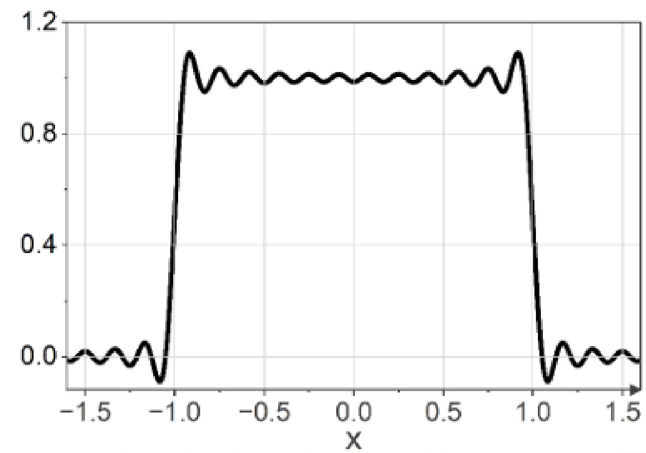
Python Implementation

```
import matplotlib.pyplot as plt
import numpy as np

# Fine resolution along path segments t in [0, 4]
t_pts = np.linspace(0, 4, 400)
work = np.zeros_like(t_pts)

# Calculate cumulative work piecewise along each segment
for i, t in enumerate(t_pts):
    if t <= 1:  # Segment O -> P
        work[i] = 4.5 * (t**2)
    elif t <= 2:  # Segment P -> Q
        tau = t - 1
        work[i] = 4.5 + (3 * tau + 0.5 * tau**2)
    elif t <= 3:  # Segment Q -> R
        tau = t - 2
        work[i] = 8.0 + (-12 * tau + 4.5 * tau**2)
    else:  # Segment R -> O
        tau = t - 3
        work[i] = 0.5 + (0.5 * tau**2 - tau)

# Initialize single figure plot
plt.figure(figsize=(8, 6))

# Plot line integral accumulation
plt.plot(t_pts, work, color="crimson", lw=2.5)
plt.axhline(0, color="black", linestyle="--", lw=1)

# Annotate key vertices along path
key_t = [0, 1, 2, 3, 4]
key_work = [0.0, 4.5, 8.0, 0.5, 0.0]
key_labels = [
    "O\n(0.0)",
    "P\n(+4.5)",
    "Q (Peak)\n(+8.0)",
    "R\n(+0.5)",
    "O (Return)\n(0.0)",
]

plt.scatter(key_t, key_work, color="darkred", s=50, zorder=5)

for kt, kw, lbl in zip(key_t, key_work, key_labels):
    plt.text(kt, kw + 0.4, lbl, ha="center", fontsize=9, fontweight="bold")

# Add text box indicating final net result
plt.annotate(
    "Net Loop Work = (+8.0) + (-8.0) = 0",
    xy=(2, 8.0),
    xytext=(1.2, 3.5),
    arrowprops=dict(facecolor="black", shrink=0.05, width=1, headwidth=5),
    fontsize=10,
    bbox=dict(boxstyle="round,pad=0.3", facecolor="yellow", alpha=0.3),
)

plt.title(
    r"Cumulative Line Integral $\int_0^t \vec{F} \cdot d\vec{l}$ Around Loop",
    fontsize=12,
)
plt.xlabel("Path Segment Sequence")
plt.ylabel("Accumulated Line Integral (Work)")
plt.xticks(key_t, ["O (0,0)", "P (1,1)", "Q (0,2)", "R (-1,1)", "O (0,0)"])
plt.ylim(-1, 10)
plt.grid(True, linestyle="--", alpha=0.4)

plt.tight_layout()
plt.show()
```



Visualising how the integrals cancel out

5 The Fourier transform

Q.12	The Fourier series representation of a square wave is shown in the figure below. The fluctuations seen near x $= \pm 1$ are named after which one of the following scientists? 
(A)	Cauchy
(B)	Fourier
(C)	Gibbs
(D)	Laplace

5.1 Problem on evaluating an integral and verifying

Problem Statement:
Consider $x(t)$ to be defined as

$$x(t) = \begin{cases} 1, & t \in (-1, 1) \\ 0, & \text{otherwise} \end{cases}$$

Consider

$$c_k = f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t) e^{-j2\pi k f_0 t}\, dt$$

Taking frequency $f_0 = \frac{1}{4}$, solve for c_k.

Step-by-Step Solution:

Step 1: Substitute $f_0 = \frac{1}{4}$ into the Integral Expression

Given $f_0 = \frac{1}{4}$, the upper and lower limits of integration become:

$$\frac{1}{2f_0} = \frac{1}{2(1/4)} = 2, \quad -\frac{1}{2f_0} = -2$$

Substituting $f_0 = \frac{1}{4}$ into c_k:

$$c_k = \frac{1}{4} \int_{-2}^{2} x(t) e^{-j\left(\frac{\pi k t}{2}\right)}\, dt$$

Step 2: Substitute $x(t)$ into the Integral

Since $x(t) = 1$ for $t \in (-1, 1)$ and 0 otherwise, the limits of integration reduce to $[-1, 1]$:

$$c_k = \frac{1}{4} \int_{-1}^{1} e^{-j\left(\frac{\pi k t}{2}\right)}\, dt$$

Step 3: Expand the Exponential using Euler's Formula

Using Euler's formula, $e^{j\theta} = \cos\theta + j\sin\theta$, we write:

$$c_k = \frac{1}{4} \int_{-1}^{1} \left[\cos\left(\frac{\pi k t}{2}\right) - j\sin\left(\frac{\pi k t}{2}\right)\right] dt$$

Step 4: Integrate the Trigonometric Terms

Performing the integration with respect to t:

$$c_k = \frac{1}{2\pi k} \left[\sin\left(\frac{\pi k t}{2}\right)\Big|_{-1}^{1} + j\cos\left(\frac{\pi k t}{2}\right)\Big|_{-1}^{1}\right]$$

Step 5: Evaluate at the Limits $t = 1$ and $t = -1$

Substitute the upper and lower limits:

$$c_k = \frac{1}{2\pi k} \left[\sin\left(\frac{\pi k}{2}\right) - \sin\left(-\frac{\pi k}{2}\right) + j\left(\cos\left(\frac{\pi k}{2}\right) - \cos\left(-\frac{\pi k}{2}\right)\right)\right]$$

Since $\sin(-x) = -\sin(x)$ and $\cos(-x) = \cos(x)$:

$$c_k = \frac{1}{2\pi k} \left[\sin\left(\frac{\pi k}{2}\right) + \sin\left(\frac{\pi k}{2}\right) + j(0)\right]$$

Step 6: Simplify to Obtain the Final Result

Combining like terms:

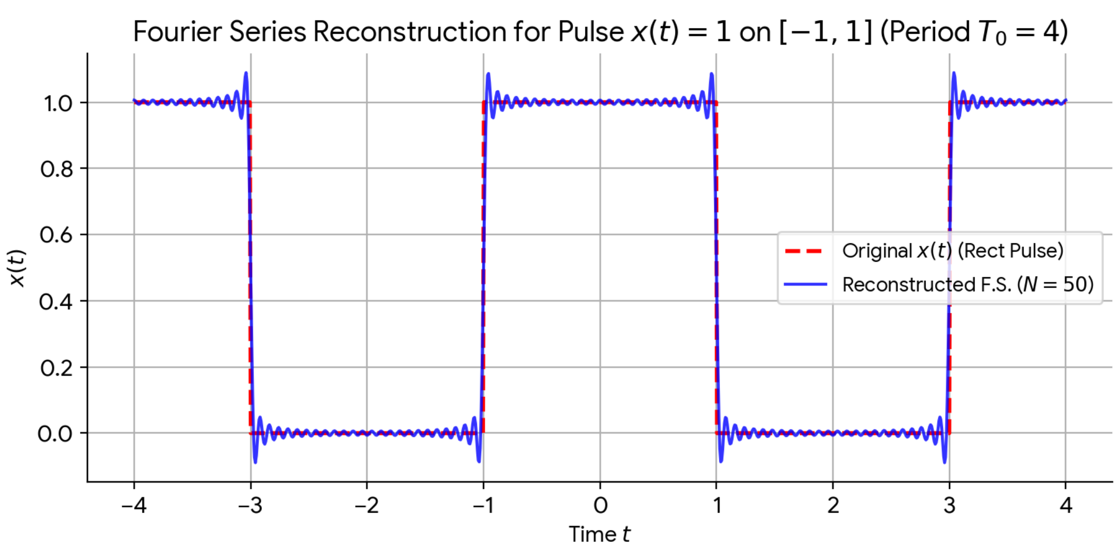
$$c_k = \frac{1}{2\pi k} \left[2\sin\left(\frac{\pi k}{2}\right)\right]$$

$$\mathbf{c_k} = \frac{\sin\left(\frac{\pi \mathbf{k}}{\mathbf{2}}\right)}{\pi \mathbf{k}}$$

Python Codes

Python Implementation

```python
import numpy as np
import matplotlib.pyplot as plt

# Parameters
T0 = 2.0          # Period T0 = 2 (interval -1 to 1)
f0 = 1.0 / T0     # Fundamental frequency f0 = 0.5 Hz
N = 50           # Number of harmonics


T0 = 4.0
f0 = 1 / T0
t = np.linspace(-T0, T0, 1000)

def compute_ck(k):
    if k == 0:
        return 0.5
    else:
        return np.sin(k * np.pi / 2) / (k * np.pi)

x_reconstructed = np.zeros_like(t, dtype=complex)
for k in range(-N, N + 1):
    c_k = compute_ck(k)
    x_reconstructed += c_k * np.exp(1j * 2 * np.pi * k * f0 * t)

x_reconstructed = np.real(x_reconstructed)

plt.figure(figsize=(8, 4))
# True x(t)
x_true = np.where(np.abs((t + T0/2) % T0 - T0/2) <= 1.0, 1.0, 0.0)
plt.plot(t, x_true, 'r--', label='Original $x(t)$ (Rect Pulse)', linewidth=2)
plt.plot(t, x_reconstructed, 'b-', label=f'Reconstructed F.S. ($N={N}$)', alpha=0.8)
plt.xlabel('Time $t$')
plt.ylabel('$x(t)$')
plt.title('Fourier Series Reconstruction for Pulse $x(t)=1$ on $[-1, 1]$ (Period $T_0=4$)')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```



Fourier series reconstruction

5.2 The Gibbs Phenomenon

Gibbs phenomenon is the overshoot and high-frequency oscillation ("ringing") that occurs when approximating a function with jump discontinuities using a finite (truncated) Fourier series.Even as the number of terms N approaches infinity, the overshoot near the jump discontinuity does not vanish; instead, it narrows in width while maintaining a fixed percentage height relative to the size of the jump.

Reason: When approximating a step function or square wave, Fourier series synthesize the shape using smooth, continuous sinusoids ($\sin$ and $\cos$). Because continuous functions cannot instantaneously step from one value to another, the Fourier series overcompensates near the vertical edge.

6 Solving for value of determinant

Q.22	An $n \times n$ square matrix A satisfies $A^T = A^{-1}$. The determinant of this matrix may take which of the following value(s)?
(A)	$+1$
(B)	-1
(C)	n
(D)	0

Problem Statement:
An $n \times n$ square matrix A satisfies $A^T = A^{-1}$. The determinant of this matrix may take which of the following value(s)?

Step-by-Step Solution:

Step 1: We are given that matrix A satisfies:

$$A^T = A^{-1}$$

Step 2: Taking the determinant of both sides of the given matrix equation:

$$\det(A^T) = \det(A^{-1})$$

Step 3: Recall the fundamental properties of matrix determinants:

- Transpose property: $\det(A^T) = \det(A)$
- Inverse property: $\det(A^{-1}) = \frac{1}{\det(A)}$

Step 4: Replacing both sides using the above properties gives:

$$\det(A) = \frac{1}{\det(A)}$$

Step 5: Multiplying both sides by $\det(A)$:

$$(\det(A))^2 = 1 \implies \det(A) = \pm 1$$

Correct Options: (A) $+1$ and **(B)** -1

Proof that $\det(A^T) = \det(A)$

When expanding $\det(A)$ along **Row 1**, you select entries across the top row and compute the determinants of their remaining submatrices.

When expanding $\det(A^T)$ down **Column 1**, those exact same entries now sit vertically in the first column, and their corresponding submatrices are simply transposed!

Visual Matrix Comparison

Matrix A (Expand along Row 1):

$$\det(A) = \begin{vmatrix} \mathbf{a_{11}} & \mathbf{a_{12}} & \mathbf{a_{13}} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Matrix A^T (Expand down Column 1):

$$\det(A^T) = \begin{vmatrix} \mathbf{a_{11}} & a_{21} & a_{31} \\ \mathbf{a_{12}} & a_{22} & a_{32} \\ \mathbf{a_{13}} & a_{23} & a_{33} \end{vmatrix}$$

Submatrix Comparison for Entry a_{12}

When selecting a_{12}:

- In A (Row 1, Column 2 deletion):

$$M_{12} = \begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}$$

- In A^T (Row 2, Column 1 deletion):

$$N_{21} = \begin{bmatrix} a_{21} & a_{31} \\ a_{23} & a_{33} \end{bmatrix}$$

Notice that $N_{21} = (M_{12})^T$. Since $\det(M_{12}) = \det((M_{12})^T)$, every term in the expansion of $\det(A)$ is identically equal to the corresponding term in $\det(A^T)$.

7 Problem on rank of a matrix and echeleon form

Q.32	If a matrix can be written as $A = uv^T$, where both u and v are n-dimensional real-valued non-zero column vectors, then the rank of the matrix A is ________ (*answer in integer*).

Problem Statement:
If a matrix can be written as $A = uv^T$, where both u and v are n-dimensional real-valued non-zero column vectors, then the rank of the matrix A is ______.

Step-by-Step solution:

Step 1: The matrix A is formed by taking the outer product of two $n \times 1$ non-zero column vectors u and v:

$$A = uv^T$$

Writing u and v^T explicitly:

$$u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}, \quad v^T = \begin{bmatrix} v_1 & v_2 & \dots & v_n \end{bmatrix}$$

Step 2: Multiplying u and v^T gives the $n \times n$ matrix A:

$$A = \begin{bmatrix} u_1v_1 & u_1v_2 & \dots & u_1v_n \\ u_2v_1 & u_2v_2 & \dots & u_2v_n \\ \vdots & \vdots & \ddots & \vdots \\ u_nv_1 & u_nv_2 & \dots & u_nv_n \end{bmatrix}$$

Step 3: Notice that each column of A can be written as a scalar multiple of vector u:

- 1st column: v_1u
- 2nd column: v_2u
- j^{th} column: v_ju

Step 4:

- The rank of a matrix is equal to the number of non zero rows in its echeleon form.
- When we reduce the above matrix into the echeleon form, we notice that each row is a multiple of another as shown above.
- Therefore, when we apply elementary row transformations by multiplying the rows with the necessary constants and subtracting, all rows but one become non zero.
- Hence, the rank of the matrix become 1.

Final Answer: 1

An example for this:

```
=== Step 1: Initial Matrix A = u * v^T ===
[ 2   6  -4]
[-1  -3   2]
[ 3   9  -6]

=============================================

=== Step 2: Elementary Row Transformations ===

Operation: R_2 -> R_2 - (-1/2) * R_1
[ 2   6  -4]
[ 0   0   0]
[ 3   9  -6]

Operation: R_3 -> R_3 - (3/2) * R_1
[ 2   6  -4]
[ 0   0   0]
[ 0   0   0]

=============================================

=== Step 3: Final Row Echelon Form ===
[ 2   6  -4]
[ 0   0   0]
[ 0   0   0]

Number of non-zero rows (Rank) = 1
```

8 Trajectory of satellite

Q.36	An earth satellite has the instantaneous position vector $\vec{r}$ and velocity vector $\vec{v}$ as given below. Here $\hat{p}$ and $\hat{q}$ denote the unit vectors along the x and y axes of the perifocal frame, respectively. Assume that the value of gravitational parameter is 398600 km^3/s^2. Which one of the following trajectories does the satellite follow? $$\vec{r} = (8000\hat{p} + 9000\hat{q})\ \text{km and } \vec{v} = (-6\hat{p} + 6\hat{q})\ \text{km/s}$$
(A)	Circle
(B)	Hyperbola
(C)	Parabola
(D)	Straight line

Problem Statement: Given the velocity and position vectors of a satellite along with its gravitational parameter $\mu = GM$, determine the trajectory of the satellite.
Given Data:

- Position vector: $\vec{r} = 8000\hat{p} + 9000\hat{q}$ km
- Velocity vector: $\vec{v} = -6\hat{p} + 6\hat{q}$ km/s
- Gravitational parameter: $\mu = 398600\ \text{km}^3/\text{s}^2$

Step-by-Step Solution:

Step 1: Calculate Magnitudes r and v^2

The magnitude of the position vector r is:

$$r = |\vec{r}| = \sqrt{8000^2 + 9000^2} = \sqrt{64 \times 10^6 + 81 \times 10^6} = \sqrt{145 \times 10^6} \approx 12041.59\ \text{km}$$

The square of the velocity magnitude v^2 is:

$$v^2 = |\vec{v}|^2 = (-6)^2 + (6)^2 = 36 + 36 = 72\ (\text{km/s})^2$$

Step 2: Calculate Specific Mechanical Energy (E)

The specific mechanical energy (energy per unit mass) is given by:

$$E = \text{Kinetic Energy per unit mass} + \text{Potential Energy per unit mass}$$

$$E = \frac{1}{2}v^2 - \frac{\mu}{r}$$

Substituting the calculated values:

$$\begin{aligned} E &= \frac{72}{2} - \frac{398600}{12041.59} \\ &= 36 - 33.102 \\ &= +2.898 \text{ km}^2/\text{s}^2 \end{aligned}$$

Step 3: Determine Trajectory Type

The trajectory of an orbiting body is determined by the sign of its specific mechanical energy E:

- $E < 0 \implies$ Elliptic / Circular orbit (Bound)
- $E = 0 \implies$ Parabolic trajectory (Unbound)
- $E > 0 \implies$ Hyperbolic trajectory (Unbound)

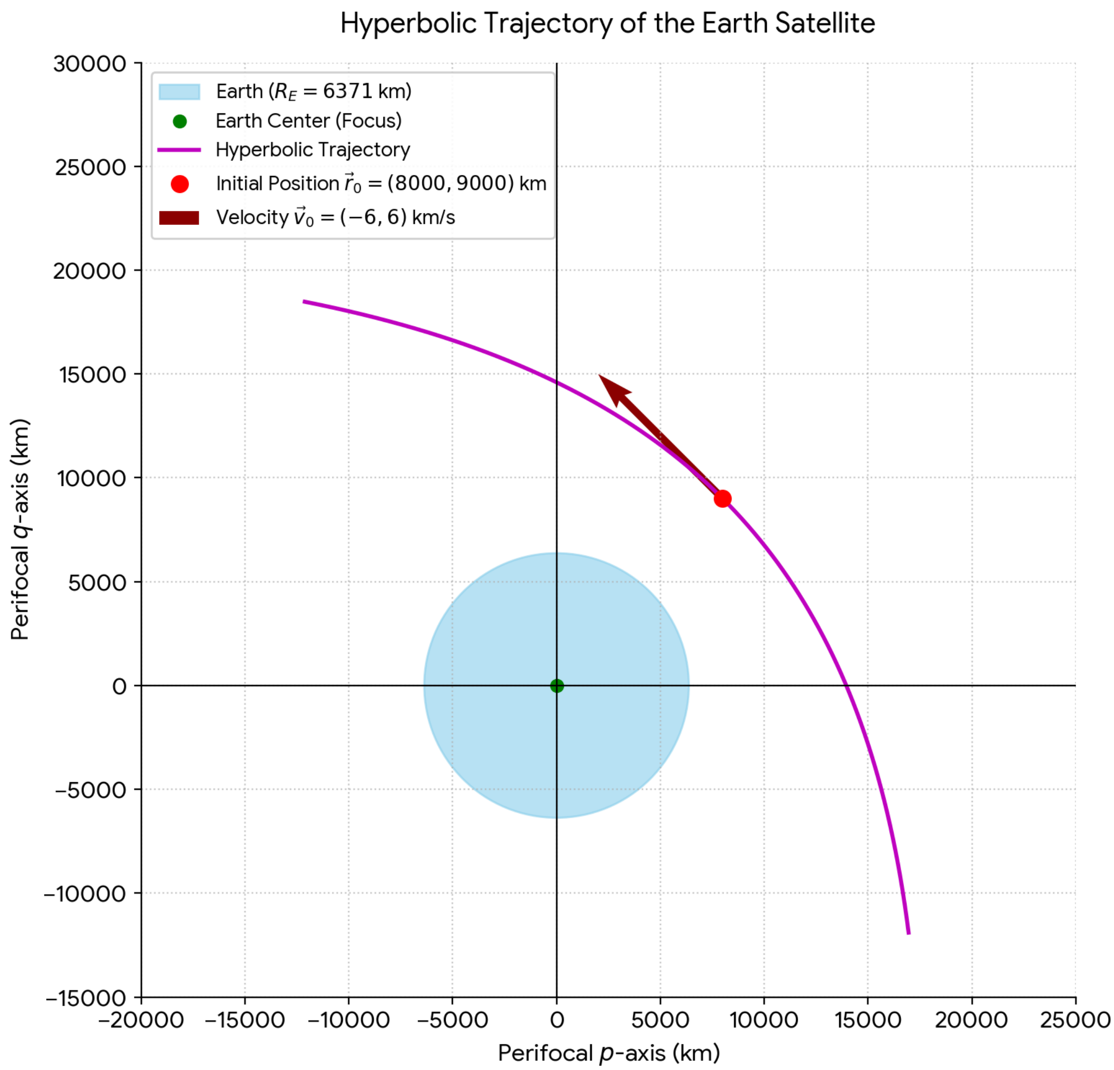
Since $E = +2.898 > 0$, the satellite follows a **Hyperbolic trajectory**.

Correct Answer: (B) Hyperbola

Python Codes

Python Implementation

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

# Constants and initial conditions
mu = 398600  # Gravitational parameter in km^3/s^2
r0 = np.array([8000.0, 9000.0])  # Position in km
v0 = np.array([-6.0, 6.0])       # Velocity in km/s

# Differential equation for two-body orbital motion: d^2r/dt^2 = -mu * r / |r|^3
def orbit_ode(t, state):
    rx, ry, vx, vy = state
    r_mag = np.sqrt(rx**2 + ry**2)
    ax = -mu * rx / (r_mag**3)
    ay = -mu * ry / (r_mag**3)
    return [vx, vy, ax, ay]

# State vector [x, y, vx, vy]
state0 = [r0[0], r0[1], v0[0], v0[1]]

# Integrate forwards and backwards in time (seconds)
t_forward = np.linspace(0, 3000, 1000)
t_backward = np.linspace(0, -3000, 1000)

sol_fwd = solve_ivp(orbit_ode, [0, 3000], state0, t_eval=t_forward, rtol=1e-9, atol=1e-9)
sol_bwd = solve_ivp(orbit_ode, [0, -3000], state0, t_eval=t_backward, rtol=1e-9, atol=1e-9)

# Combine forward and backward trajectories
x_traj = np.concatenate((sol_bwd.y[0][::-1], sol_fwd.y[0]))
y_traj = np.concatenate((sol_bwd.y[1][::-1], sol_fwd.y[1]))

fig, ax = plt.subplots(figsize=(8, 8))

# Draw Earth at the focus (origin)
earth_radius = 6371  # Mean radius in km
earth = plt.Circle((0, 0), earth_radius, color='skyblue', alpha=0.6, label='Earth ($R_E = 6371$ km)')
ax.add_patch(earth)
ax.plot(0, 0, 'go', markersize=6, label='Earth Center (Focus)')

# Draw Hyperbolic Trajectory
ax.plot(x_traj, y_traj, 'm-', linewidth=2, label='Hyperbolic Trajectory')

# Mark Initial Position & Velocity Vector
ax.plot(r0[0], r0[1], 'ro', markersize=8, label=r'Initial Position $\vec{r}_0 = (8000, 9000)$ km')
ax.quiver(r0[0], r0[1], v0[0], v0[1], angles='xy', scale_units='xy', scale=0.001,
          color='darkred', width=0.008, label=r'Velocity $\vec{v}_0 = (-6, 6)$ km/s')

# Labels and formatting
ax.set_title('Hyperbolic Trajectory of the Earth Satellite', fontsize=14, pad=15)
ax.set_xlabel('Perifocal $p$-axis (km)', fontsize=12)
ax.set_ylabel('Perifocal $q$-axis (km)', fontsize=12)
ax.grid(True, linestyle=':', alpha=0.7)
ax.axhline(0, color='black', linewidth=0.8)
ax.axvline(0, color='black', linewidth=0.8)
ax.set_aspect('equal', adjustable='box')
ax.set_xlim(-20000, 25000)
ax.set_ylim(-15000, 30000)
ax.legend(loc='upper left', fontsize=10, framealpha=0.9)

plt.tight_layout()
plt.show()
```



Trajectory generated using python

9 Problem on eigenvectors and eigenvalues

Q.37	For the matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, if the relation $a + b = c + d$ holds and $a, b, c, d \neq 0$, then which one of the following statements about A is FALSE?
(A)	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector
(B)	$\lambda = a + b$ is an eigenvalue
(C)	$\lambda = d - b$ is an eigenvalue
(D)	$\lambda = d + b$ is an eigenvalue

Problem Statement: For the matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, if the relation $a + b = c + d$ holds and $a, b, c, d \neq 0$, then which one of the following statements about A is FALSE?

(A) $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector

(B) $\lambda = a + b$ is an eigenvalue

(C) $\lambda = d - b$ is an eigenvalue

(D) $\lambda = d + b$ is an eigenvalue

Step-by-Step Solution

Step 1: Check Eigenvector and First Eigenvalue
Given that $a + b = c + d = k$:

$$\begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = k = \begin{pmatrix} c & d \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\implies \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} k \\ k \end{pmatrix}$$

$$\implies \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = k \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$\therefore \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an eigenvector and $k = a + b$ is an eigenvalue.

Step 2: Find the Second Eigenvalue
From the given relation $a + b = c + d$:

$$a - c = d - b = -(b - d) = k'$$

Evaluating the matrix-vector multiplication:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} a - b \\ c - d \end{pmatrix} = \begin{pmatrix} k' \\ -k' \end{pmatrix} = k' \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$\therefore \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is an eigenvector and $d - b$ is the 2^{nd} eigenvalue.

Conclusion:

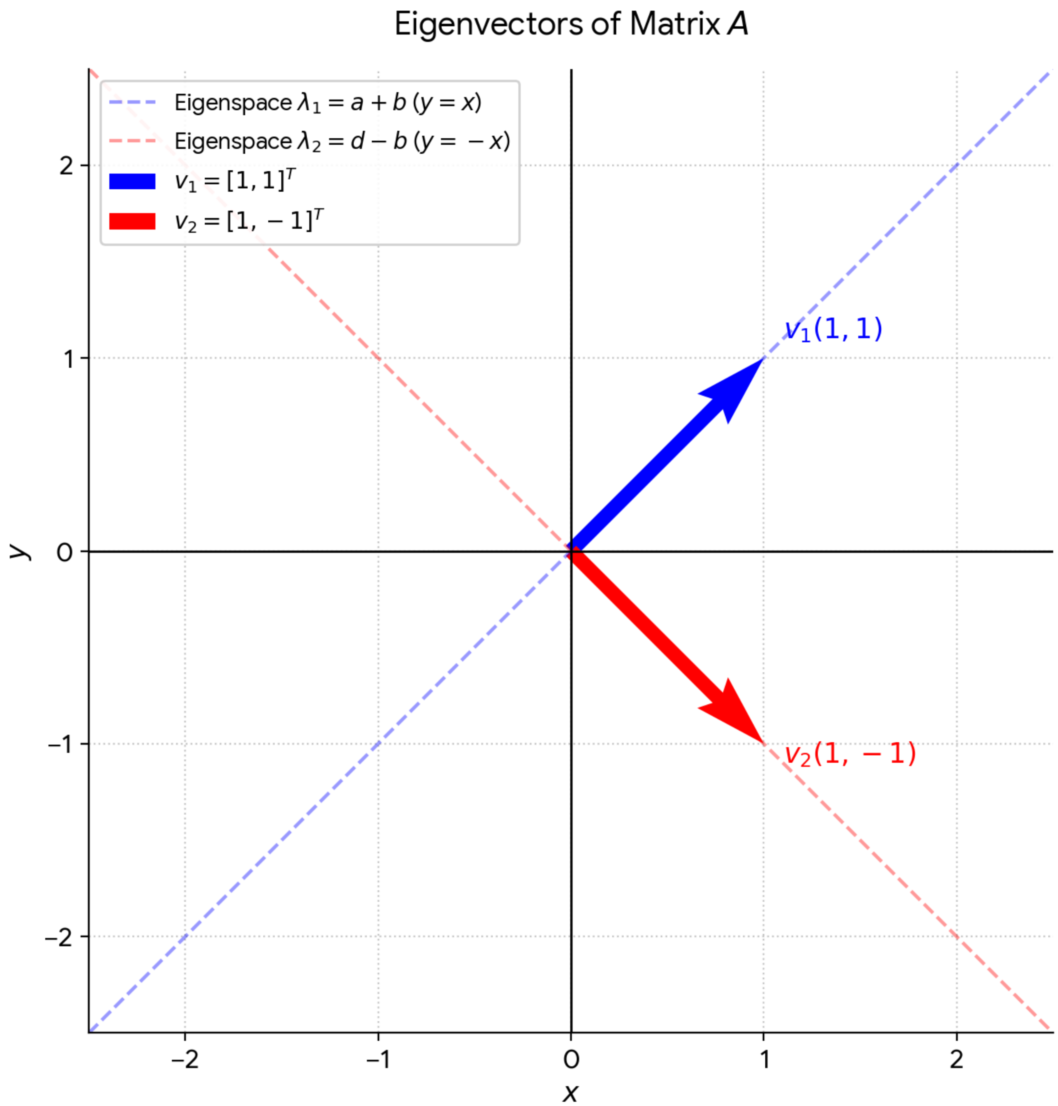
- Statement (A) is **TRUE**: $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector.
- Statement (B) is **TRUE**: $\lambda = a + b$ is an eigenvalue.
- Statement (C) is **TRUE**: $\lambda = d - b$ is an eigenvalue.
- Statement (D) is **FALSE**: $\lambda = d + b$ is not an eigenvalue.

Correct Answer: (D)

Python Codes

Python Implementation

```python
import numpy as np
import matplotlib.pyplot as plt

# Define the eigenvectors
v1 = np.array([1, 1])
v2 = np.array([1, -1])

# Origin (0, 0)
origin = np.array([0, 0])

# Create figure
fig, ax = plt.subplots(figsize=(7, 7))

# Plot Eigenspaces (lines spanned by eigenvectors)
x_span = np.linspace(-2.5, 2.5, 100)
ax.plot(x_span, x_span, 'b--', alpha=0.4, label=r'Eigenspace $\lambda_1 = a+b$ ($y = x$)')
ax.plot(x_span, -x_span, 'r--', alpha=0.4, label=r'Eigenspace $\lambda_2 = d-b$ ($y = -x$)')

# Plot Eigenvectors as arrows
ax.quiver(*origin, *v1, angles='xy', scale_units='xy', scale=1,
          color='blue', width=0.015, label=r'$v_1 = [1, 1]^T$')
ax.quiver(*origin, *v2, angles='xy', scale_units='xy', scale=1,
          color='red', width=0.015, label=r'$v_2 = [1, -1]^T$')

# Annotate vectors directly on the plot
ax.text(1.1, 1.1, r'$v_1(1, 1)$', fontsize=12, color='blue', fontweight='bold')
ax.text(1.1, -1.1, r'$v_2(1, -1)$', fontsize=12, color='red', fontweight='bold')

# Configure grid and axes
ax.set_xlim(-2.5, 2.5)
ax.set_ylim(-2.5, 2.5)
ax.axhline(0, color='black', linewidth=1)
ax.axvline(0, color='black', linewidth=1)
ax.grid(True, linestyle=':', alpha=0.7)
ax.set_aspect('equal', adjustable='box')

# Labels and Title
ax.set_title('Eigenvectors of Matrix $A$', fontsize=14, pad=15)
ax.set_xlabel('$x$', fontsize=12)
ax.set_ylabel('$y$', fontsize=12)
ax.legend(loc='upper left', framealpha=0.9)

# Display plot
plt.tight_layout()
plt.show()
```



Plot generated through python, showing the eigen vectors

10 Differential equation using laplace transform

Q.49	Consider the differential equation with the initial conditions given below. If $y(x)$ is the solution of the equation, the value of the slope, $\dfrac{dy}{dx}$, at $x = \ln(2)$ is _________ (*rounded off to three decimal places*). $$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0 \quad \text{with } y|_{x=0} = 0 \text{ and } \left.\frac{dy}{dx}\right|_{x=0} = 1$$

Problem Statement:
Consider the differential equation:

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$$

with initial conditions $y|_{x=0} = 0$ and $\left.\frac{dy}{dx}\right|_{x=0} = 1$. Find the value of the slope $\frac{dy}{dx}$ at $x = \ln(2)$ rounded off to three decimal places.

Step-by-Step Solution

Step 1: Apply Laplace Transform

Let $Y(s) = \mathcal{L}\{y(x)\}$. Taking the Laplace transform on both sides:

$$\mathcal{L}\left\{\frac{d^2y}{dx^2}\right\} + 2\mathcal{L}\left\{\frac{dy}{dx}\right\} + \mathcal{L}\{y\} = 0$$

Using the standard operational properties:

$$\left[s^2Y(s) - sy(0) - y'(0)\right] + 2\left[sY(s) - y(0)\right] + Y(s) = 0$$

Substituting the given initial conditions $y(0) = 0$ and $y'(0) = 1$:

$$(s^2Y(s) - 1) + 2sY(s) + Y(s) = 0$$

$$(s^2 + 2s + 1)Y(s) = 1$$

$$Y(s) = \frac{1}{(s+1)^2}$$

Step 2: Inverse Laplace Transform

Taking the inverse Laplace transform gives the solution $y(x)$:

$$y(x) = \mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2}\right\} = xe^{-x}$$

Step 3: Calculate the Slope $\frac{dy}{dx}$

Differentiating $y(x)$ with respect to x:

$$\frac{dy}{dx} = \frac{d}{dx}\left(xe^{-x}\right) = (1-x)e^{-x}$$

Step 4: Evaluate at $x = \ln(2)$

Substituting $x = \ln(2)$:

$$\left.\frac{dy}{dx}\right|_{x=\ln(2)} = \big(1-\ln(2)\big)e^{-\ln(2)}$$

Since $e^{-\ln(2)} = \frac{1}{2}$:

$$\left.\frac{dy}{dx}\right|_{x=\ln(2)} = \frac{1-\ln(2)}{2} \approx \frac{1-0.693147}{2} = 0.153426$$

Rounding off to three decimal places yields: **0.153**

11 Sub-Problems on laplace transform

11.1 Few proofs on laplace transform

Consider a function $g(t)$ and let $G(s)$ be its Laplace transform. Show that:

(a) Applying Laplace transform on $g'(t)$ gives $sG(s)$

(b) Applying Laplace transform on $g''(t)$ gives $s^2G(s)$

Step-by-Step Solution & Proof
By definition, the Laplace transform of a function $f(t)$ is given by:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt = F(s)$$

Given that $\mathcal{L}\{g(t)\} = G(s) = \int_0^\infty e^{-st} g(t)\, dt$, and assuming zero initial conditions $g(0) = 0$ and $g'(0) = 0$:

(a) Proof for $g'(t)$

Using the definition of the Laplace transform:

$$\mathcal{L}\{g'(t)\} = \int_0^\infty e^{-st} g'(t)\, dt$$

Applying integration by parts with $u = e^{-st}$ and $dv = g'(t)\, dt$ (giving $du = -se^{-st}\, dt$ and $v = g(t)$):

$$\begin{aligned}\mathcal{L}\{g'(t)\} &= \left[e^{-st} g(t)\right]_0^\infty - \int_0^\infty (-se^{-st}) g(t)\, dt \\ &= \left(\lim_{t\to\infty} e^{-st} g(t) - g(0)\right) + s\int_0^\infty e^{-st} g(t)\, dt\end{aligned}$$

Assuming $g(t)$ is of exponential order such that $\lim_{t\to\infty} e^{-st} g(t) = 0$, and setting $g(0) = 0$:

$$\mathcal{L}\{g'(t)\} = 0 - 0 + sG(s) = sG(s)$$

(b) Proof for $g''(t)$

Using the definition for $g''(t)$:

$$\mathcal{L}\{g''(t)\} = \int_0^\infty e^{-st} g''(t)\, dt$$

Applying integration by parts with $u = e^{-st}$ and $dv = g''(t)\, dt$ (giving $du = -se^{-st}\, dt$ and $v = g'(t)$):

$$\begin{aligned}\mathcal{L}\{g''(t)\} &= \left[e^{-st} g'(t)\right]_0^\infty - \int_0^\infty (-se^{-st}) g'(t)\, dt \\ &= \left(\lim_{t\to\infty} e^{-st} g'(t) - g'(0)\right) + s\int_0^\infty e^{-st} g'(t)\, dt\end{aligned}$$

With $g'(0) = 0$ and substituting the result from part (a), $\mathcal{L}\{g'(t)\} = sG(s)$:

$$\mathcal{L}\{g''(t)\} = 0 - 0 + s\big(sG(s)\big) = s^2G(s)$$

11.2 Problem on laplace transform

Problem Statement:
Consider a function $u(t)$ as defined below:

$$u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \\ \frac{1}{2}, & t = 0 \end{cases}$$

Apply the Laplace transform on $u(t)$.

Step-by-Step Solution By definition, the unilateral Laplace transform of a function $f(t)$ is given by:

$$\mathcal{L}\{f(t)\} = \int_0^\infty f(t)e^{-st}\,dt$$

Step 1: Set up the Integral

For $t > 0$, the function takes the value $u(t) = 1$. Because altering a function's value at a single point ($t = 0$) does not affect the value of a definite integral:

$$\mathcal{L}\{u(t)\} = \int_0^\infty (1) \cdot e^{-st}\,dt$$

Step 2: Evaluate the Integral

Integrating with respect to t:

$$\begin{aligned}
\mathcal{L}\{u(t)\} &= \left[\frac{e^{-st}}{-s}\right]_0^\infty \\
&= \lim_{t\to\infty}\left(\frac{e^{-st}}{-s}\right) - \left(\frac{e^0}{-s}\right) \\
&= 0 - \left(-\frac{1}{s}\right) \\
&= \frac{1}{s}
\end{aligned}$$

Final Answer

$$\mathcal{L}\{u(t)\} = \frac{1}{s}$$

12 The modulus function

Q.51	The minimum value of the function $f(x) = \lvert x \rvert + \lvert 2x + 3 \rvert$ for real x is ________ (*rounded off to 1 decimal place*).

Problem Statement:
Find the minimum value of the function $f(x) = |x| + |2x+3|$ for real x.

Step-by-Step Solution:

Step 1: Identify Critical Points

The critical points occur where the expressions inside the absolute values equal zero:

$$x = 0$$

$$2x + 3 = 0 \implies x = -\frac{3}{2} = -1.5$$

These two critical points divide the real line into three sub-intervals:

$$(-\infty, -1.5), \quad [-1.5, 0), \quad \text{and} \quad [0, \infty)$$

Step 2: Express $f(x)$ as a Piecewise Function

1. **For $x < -1.5$:**
   Both expressions inside the absolute values are negative, so $|x| = -x$ and $|2x+3| = -(2x+3)$:

   $$f(x) = -x - (2x+3) = -3x - 3$$

2. **For $-1.5 \le x < 0$:**
   The term x is negative ($|x| = -x$) while $2x+3$ is non-negative ($|2x+3| = 2x+3$):

   $$f(x) = -x + (2x+3) = x + 3$$

3. **For $x \ge 0$:**
   Both expressions are non-negative, so $|x| = x$ and $|2x+3| = 2x+3$:

   $$f(x) = x + (2x+3) = 3x + 3$$

Combining these cases gives the complete piecewise definition:

$$f(x) = \begin{cases} -3x - 3, & \text{if } x < -1.5 \\ x + 3, & \text{if } -1.5 \le x < 0 \\ 3x + 3, & \text{if } x \ge 0 \end{cases}$$

Step 3: Determine the Minimum Value

Analyzing the behavior of each linear segment:

- On $(-\infty, -1.5)$, the line $f(x) = -3x - 3$ has a negative slope (−3), so $f(x)$ strictly decreases as x approaches -1.5.
- On $[-1.5, 0)$, the line $f(x) = x + 3$ has a positive slope (+1), so $f(x)$ strictly increases as x moves away from -1.5.
- On $[0, \infty)$, the line $f(x) = 3x + 3$ has a positive slope (+3), so $f(x)$ continues to increase.

Since $f(x)$ decreases for $x < -1.5$ and increases for $x > -1.5$, the minimum value occurs at $x = -1.5$:

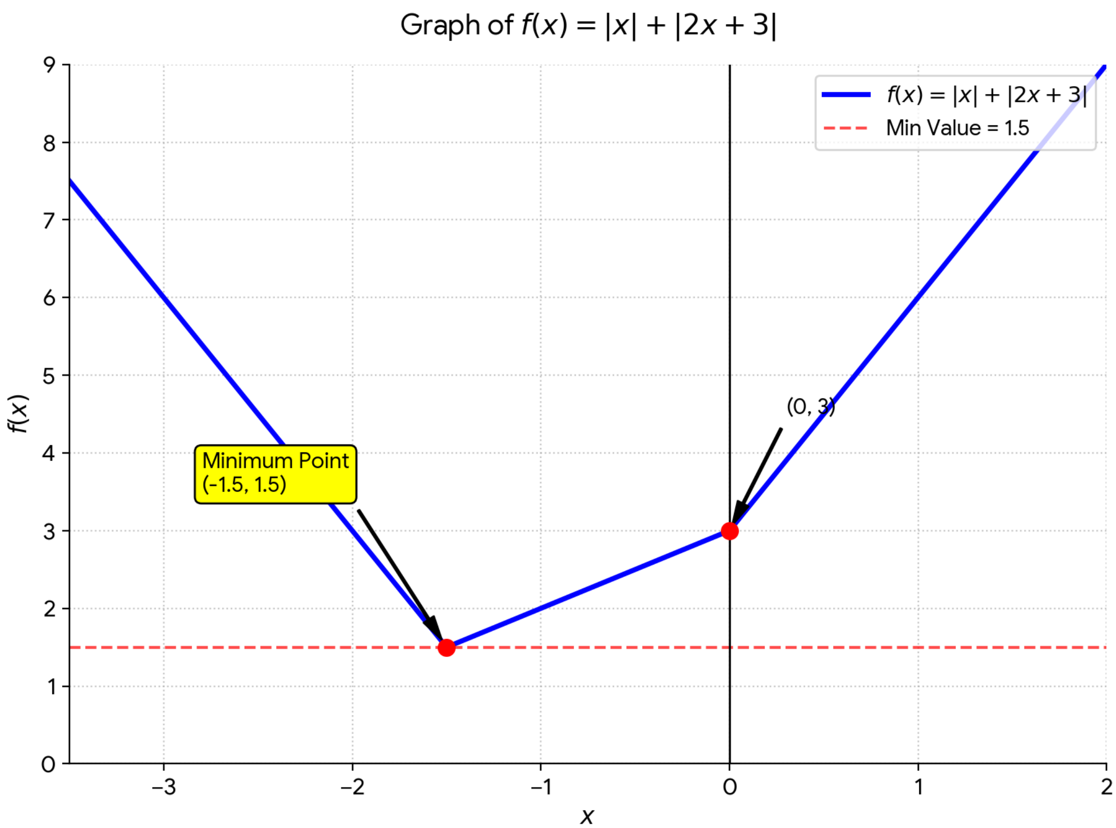
$$f(-1.5) = -1.5 + 3 = 1.5$$

Final Answer: 1.5

Python Codes

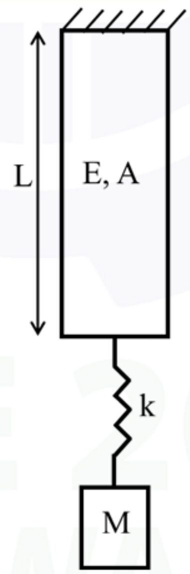
Python Implementation

```python
import numpy as np
import matplotlib.pyplot as plt

# Define x range around critical points (-1.5 and 0)
x = np.linspace(-3.5, 2.0, 500)
y = np.abs(x) + np.abs(2 * x + 3)

fig, ax = plt.subplots(figsize=(8, 6))

# Plot the function
ax.plot(x, y, 'b-', linewidth=2.5, label=r'$f(x) = |x| + |2x + 3|$')

# Highlight critical points (-1.5, 1.5) and (0, 3)
crit_x = [-1.5, 0]
crit_y = [1.5, 3.0]
ax.plot(crit_x, crit_y, 'ro', markersize=8, zorder=5)

# Annotate minimum point
ax.annotate('Minimum Point\n(-1.5, 1.5)', xy=(-1.5, 1.5), xytext=(-2.8, 3.5),
            arrowprops=dict(facecolor='black', shrink=0.05, width=1, headwidth=6),
            fontsize=11, fontweight='bold',
            bbox=dict(boxstyle="round,pad=0.3", fc="yellow", ec="black", lw=1))

# Annotate corner at x = 0
ax.annotate('(0, 3)', xy=(0, 3), xytext=(0.3, 4.5),
            arrowprops=dict(facecolor='black', shrink=0.05, width=1, headwidth=6),
            fontsize=11, fontweight='bold')

# Horizontal line at minimum value
ax.axhline(1.5, color='r', linestyle='--', alpha=0.7, label='Min Value = 1.5')

# Labels and Grid
ax.set_title(r'Graph of $f(x) = |x| + |2x + 3|$', fontsize=14, pad=15)
ax.set_xlabel('$x$', fontsize=12)
ax.set_ylabel('$f(x)$', fontsize=12)
ax.grid(True, linestyle=':', alpha=0.7)
ax.axhline(0, color='black', linewidth=1)
ax.axvline(0, color='black', linewidth=1)
ax.legend(loc='upper right', fontsize=11)

# Axis limits
ax.set_xlim(-3.5, 2.0)
ax.set_ylim(0, 9)

plt.tight_layout()
plt.show()
```



Plot of the function

13 Problem on solids and oscillations

Q.59	A system comprising a bar, spring and mass is shown in the figure below. The bar, having negligible mass, is made of a material having Young's modulus E = 200 GPa, cross-sectional area A = 100 mm^2, and length L = 100 mm. The spring stiffness k = 200 kN/mm and the mass M = 100 kg. The natural frequency of free vibration of the system is ________ rad/s (*rounded off to the nearest integer*). 

Problem Statement:

A bar, spring, and mass system is arranged in series.

- Bar: $E = 200$ GPa, $A = 100$ mm^2, $L = 100$ mm
- Spring Stiffness: $k = 200$ kN/mm
- Mass: $M = 100$ kg

Find the natural frequency of free vibration ω_n in rad/s.

Step-by-Step Solution:

Step 1: Axial Stiffness of the Bar (k_{bar})

The axial stiffness of the uniform bar is:

$$k_{\text{bar}} = \frac{AE}{L}$$

Converting parameters to standard SI units:

$$E = 200 \times 10^9 \text{ N/m}^2$$
$$A = 100 \times 10^{-6} \text{ m}^2 = 10^{-4} \text{ m}^2$$
$$L = 100 \times 10^{-3} \text{ m} = 0.1 \text{ m}$$

$$k_{\text{bar}} = \frac{10^{-4} \times 200 \times 10^9}{0.1} = 2 \times 10^8 \text{ N/m}$$

Step 2: Equivalent Stiffness (k_{eq})

The bar and the helical spring are connected in series. Therefore:

$$\frac{1}{k_{\text{eq}}} = \frac{1}{k_{\text{bar}}} + \frac{1}{k}$$

With $k = 200$ kN/mm $= 2 \times 10^8$ N/m:

$$k_{\text{eq}} = \frac{k_{\text{bar}} \cdot k}{k_{\text{bar}} + k} = \frac{(2 \times 10^8)(2 \times 10^8)}{2 \times 10^8 + 2 \times 10^8} = 10^8 \text{ N/m}$$

Step 3: Natural Frequency (ω_n)

The natural frequency of free vibration is:

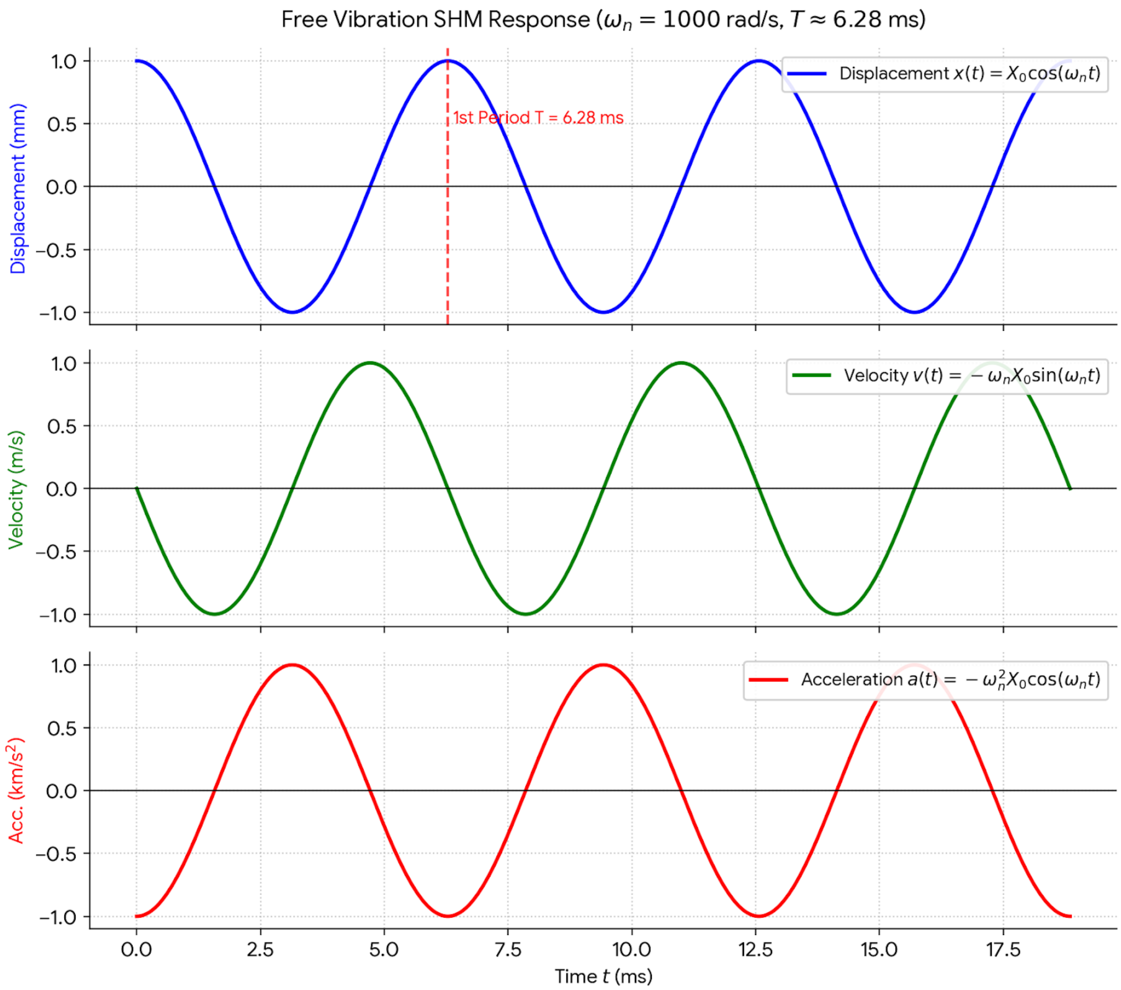
$$\omega_n = \sqrt{\frac{k_{\text{eq}}}{M}} = \sqrt{\frac{10^8}{100}} = \sqrt{10^6} = 1000 \text{ rad/s}$$

Final Answer: 1000

Python Codes

Python Implementation

```
import numpy as np
import matplotlib.pyplot as plt

# System Parameters
E = 200e9       # Pa
A = 100e-6      # m^2
L = 100e-3      # m
k_spring = 200e6 # N/m
M = 100         # kg

# Stiffness calculation
k_bar = (A * E) / L
k_eq = (k_bar * k_spring) / (k_bar + k_spring)
omega_n = np.sqrt(k_eq / M) # rad/s (1000 rad/s)

# Time period T
T = 2 * np.pi / omega_n # seconds (~0.00628 s = 6.28 ms)

# Define time array for 3 complete cycles in milliseconds
t = np.linspace(0, 3 * T, 1000) # seconds
t_ms = t * 1000                 # milliseconds

# Assume unit initial displacement X_0 = 1 mm (0.001 m)
X0 = 1.0 # mm
x_t = X0 * np.cos(omega_n * t)                   # Displacement in mm
v_t = -X0 * (omega_n / 1000) * np.sin(omega_n * t) # Velocity in m/s
a_t = -X0 * ((omega_n / 1000)**2) * np.cos(omega_n * t) # Acceleration in km/s^2

# Plot displacement, velocity, and acceleration
fig, (ax1, ax2, ax3) = plt.subplots(3, 1, figsize=(9, 8), sharex=True)

# 1. Displacement Plot
ax1.plot(t_ms, x_t, 'b-', linewidth=2, label=r'Displacement $x(t) = X_0 \cos(\omega_n t)$')
ax1.set_ylabel('Displacement (mm)', fontsize=11, color='blue')
ax1.grid(True, linestyle=':', alpha=0.7)
ax1.axhline(0, color='black', linewidth=0.8)
ax1.set_title(f'Free Vibration SHM Response ($\omega_n = {omega_n:.0f}$ rad/s, $T \\approx {T*1000:.2f}$ ms)', fontsize=13, pad=12)
ax1.legend(loc='upper right', framealpha=0.9)

# Mark time period T on first plot
ax1.axvline(T * 1000, color='red', linestyle='--', alpha=0.7)
ax1.text(T * 1000 + 0.1, 0.5, r'$1^\text{st}$ Period $T = 6.28$ ms', color='red', fontsize=10, fontweight='bold')

# 2. Velocity Plot
ax2.plot(t_ms, v_t, 'g-', linewidth=2, label=r'Velocity $v(t) = -\omega_n X_0 \sin(\omega_n t)$')
ax2.set_ylabel('Velocity (m/s)', fontsize=11, color='green')
ax2.grid(True, linestyle=':', alpha=0.7)
ax2.axhline(0, color='black', linewidth=0.8)
ax2.legend(loc='upper right', framealpha=0.9)

# 3. Acceleration Plot
ax3.plot(t_ms, a_t, 'r-', linewidth=2, label=r'Acceleration $a(t) = -\omega_n^2 X_0 \cos(\omega_n t)$')
ax3.set_xlabel('Time $t$ (ms)', fontsize=11)
ax3.set_ylabel(r'Acc. ($\text{km/s}^2$)', fontsize=11, color='red')
ax3.grid(True, linestyle=':', alpha=0.7)
ax3.axhline(0, color='black', linewidth=0.8)
ax3.legend(loc='upper right', framealpha=0.9)

plt.tight_layout()
plt.show()
```



Plot of the position, velocity and acceleration vectors of the system